

Literature Review

S.No	Paper Title	Authors	Year	Technique Used	Dataset	Accuracy	Gap / Limitation
1	Real-Time Drowsiness Detection Using Eye Aspect Ratio and Facial Landmark Detection	Prerana et al.	2024	dlib 68-point facial landmarks + Eye Aspect Ratio (EAR) threshold-based detection	Custom webcam dataset	~91%	Fixed EAR threshold (not working for people with narrow eye sockets and/or glasses). No deep learning model is directly used.
2	A Real-Time Embedded System for Driver Drowsiness Detection Based on CNN and MAR	Espinosa et al.	2024	CNN + EAR/MAR calculation using dlib landmarks on NVIDIA Jetson Nano embedded board	Custom NIR (Near Infrared) camera dataset	97.44%	Uses embedded hardware (Jetson Nano) which requires a special device. There is no web-based user interface or accessible user interface.
3	Improving Driver Drowsiness Detection via Personalised EAR/MAR Thresholds and CNN Classification	Sanchez-Gendriz et al.	2025	Personalised EAR and MAR threshold calibration combined with CNN binary classifier	Custom driving dataset	94%	Doesn't include head pose estimation. No deployment integrated application interface.
4	Driver Monitoring System Using MediaPipe and MobileNetV2 for Real-Time Fatigue Detection	Rosero-Montalvo et al.	2026	MediaPipe 468 facial landmarks + MobileNetV2 CNN + EAR + MAR + head pose estimation	27 participants in real and simulated driving conditions	88.89%	No session event logging or session analytics. There is no accessible deployment interface for the Web or mobile devices.
5	CNN-Based Eye State Classification for Drowsiness Detection Using MediaPipe	Castro-Ospina et al.	2023	Comparison of ResNet50V2, VGG16 and InceptionV3 CNNs on cropped eye region images	NITYMED video dataset	99.71%	Heavy models that are not appropriate for real-time inference on typical PC systems. No "Yawning or head pose" module.
6	Real-Time Driver Drowsiness Detection Using Vision Transformer Architectures	Jarndal et al.	2025	Vision Transformer (ViT) and Swin Transformer evaluated on eye state classification	MRL Eye Dataset	99.15%	Computational demands of the transformer models are too high for real-time use on regular hardware. No multi-cue fusion.

Research Gap Summary

Current research mainly highlights the individual detection modules, such as EAR-only, CNN-only or Head Pose-only. Most systems do not have a single multi-cue fusion pipeline. Moreover, there is no work which integrates CNN based eye state classification, eye eyes/attention recognition, MediaPipe head pose estimation and deployment platform to a web-based real-time system. The proposed project discusses all-four areas and integrates them within a session-based logging of the events and drowsiness severity score as displayed in a Streamlit web application.